Inferring Behavioural Pattern of Seabirds For Effective Conservation

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Motivation

Seabirds are

- good indicators of the marine ecosystem health
- offered less protection at sea
- in fast decline than other birds in the UK

Figure 1: *Red listed seabirds in the UK*

RESEARCH GOAL

Better behavioural inference of seabirds from tracking data for effective conservation actions and management: **identifying important at-sea areas**

PROBLEM 1

Figure 2: *GPS tracker on an Albatross*

Behavioural states of seabirds are mostly inferred using hidden Markov models (HMMs) from telemetry data in practice without validation.

WORK FLOW

- **Assess** visual tracking data* of four terns species during breeding season
- **Infer** behaviours from visual tracking data using HMMs
- **Validate** inferred behaviours using real observations

*****_Collected in the UK by the Joint Nature Conservation Committee (2009-2011)_

STUDY SITES

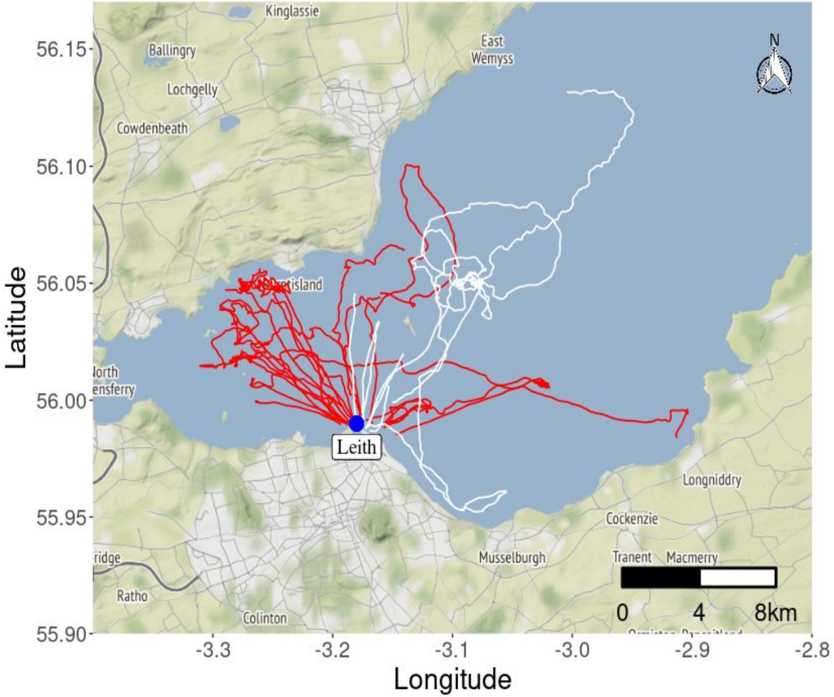
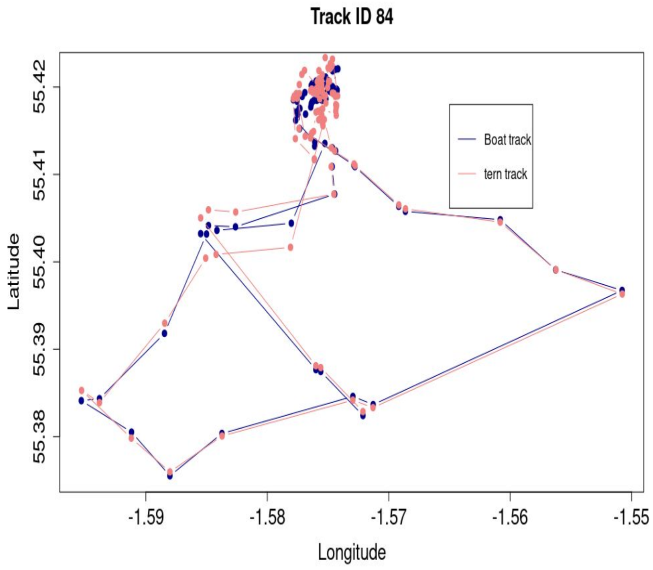


Figure 3: *Visual tracks of common tern at Leith breeding colony during chick rearing (red) tracks and incubation (white) tracks*

- Given the bearing and distance of the boat from terns, location of terns is computed using Haversine formula



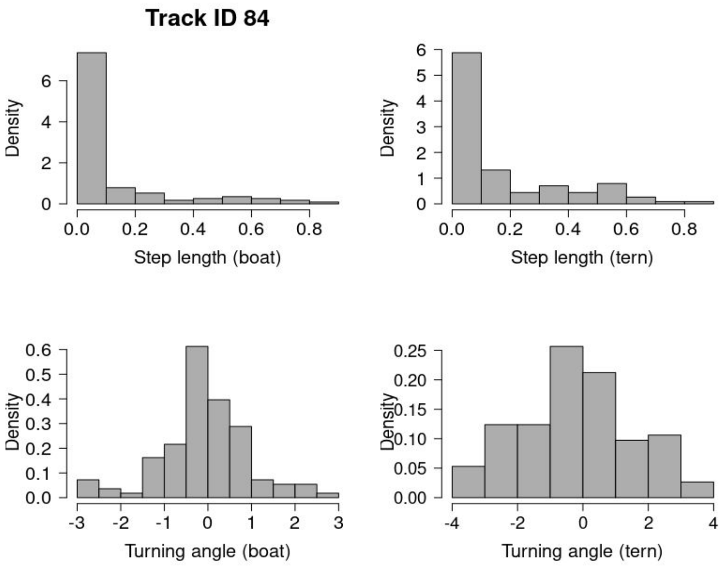


Figure 4: *right panel: visual and approximate tracking data of a Roseate tern in Coquet Island, 2009 (left panel), and their corresponding step length and turning angle distributions (right panel)*

Hidden Markov Model

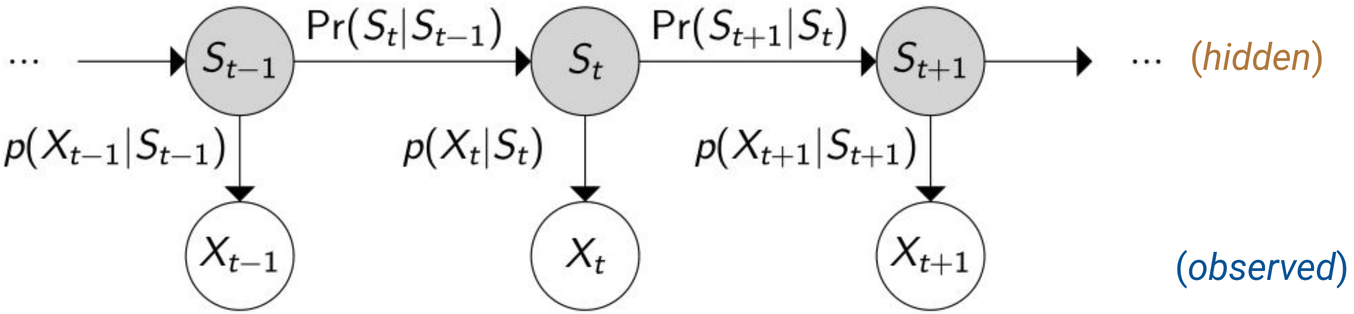


Figure 6: *Representation of a HMM*

- Step length (gamma) and turning angle (von Mises) obtained from visual tracking data of terns
- Tern behavioural states: foraging and not-foraging

ASSESSING VISUAL TRACKING DATA

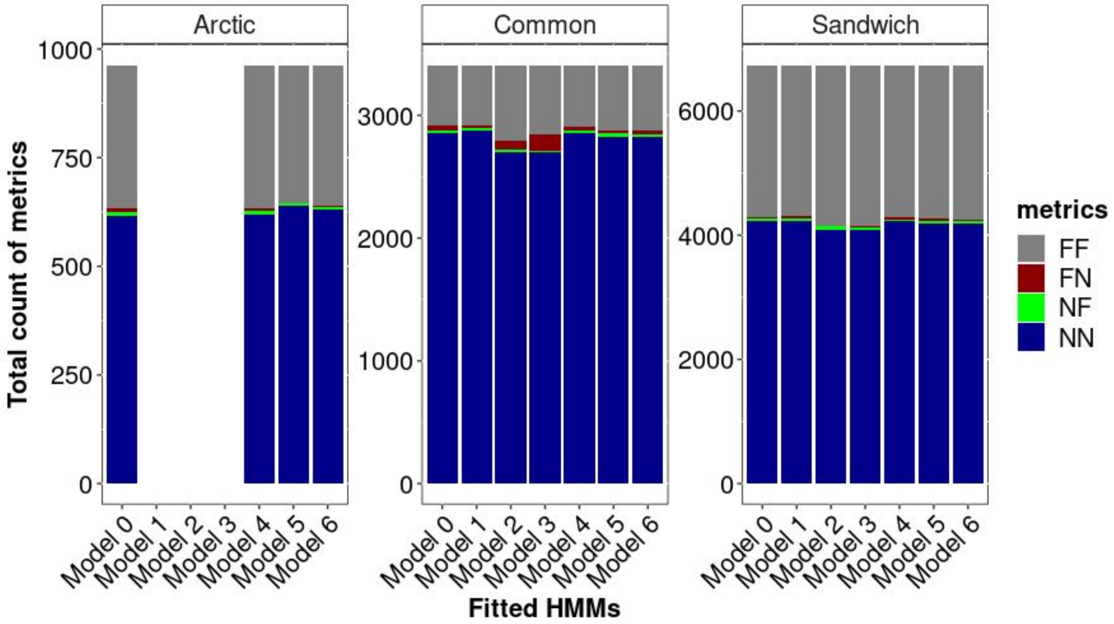


Figure 5: *Confusion matrix metrics of inferred behavioural states from HMMs fitted to visual tracking and approximate location data of 9 terns: 1 Arctic, 2 Common and 6 Sandwich, in Coquet colony during chick-rearing, 2009.*

FF = Foraging/Foraging (true positive), FN = Foraging/Non-foraging (false negative) NF = Non-foraging/Foraging (false positive), NN = Non-Foraging/Non-foraging (true negative)

Movement Metrics

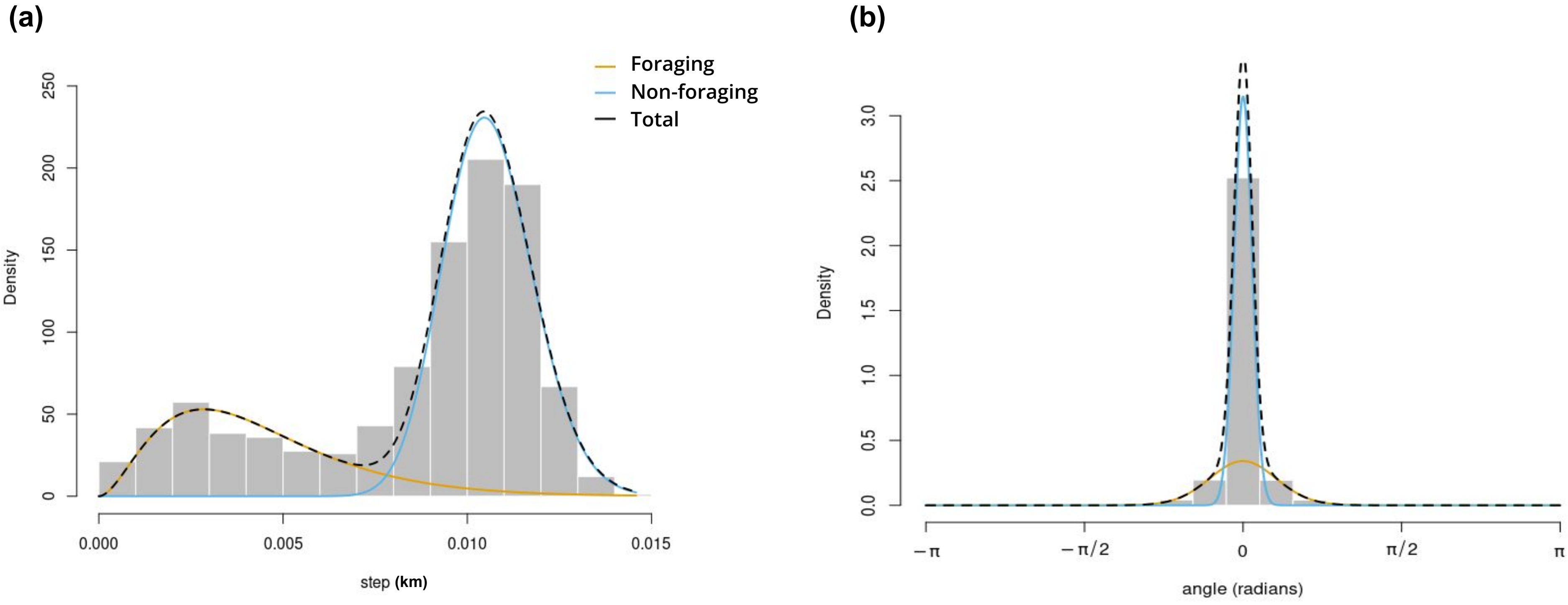


Figure 6: *HMM estimated state-dependent distributions for the step lengths (a) and the turning angles (b) of visually-tracked 6 common tern in South Shian colony during chick-rearing period, 2011.*

VALIDATING HMM-INFERRED BEHAVIOURAL STATES

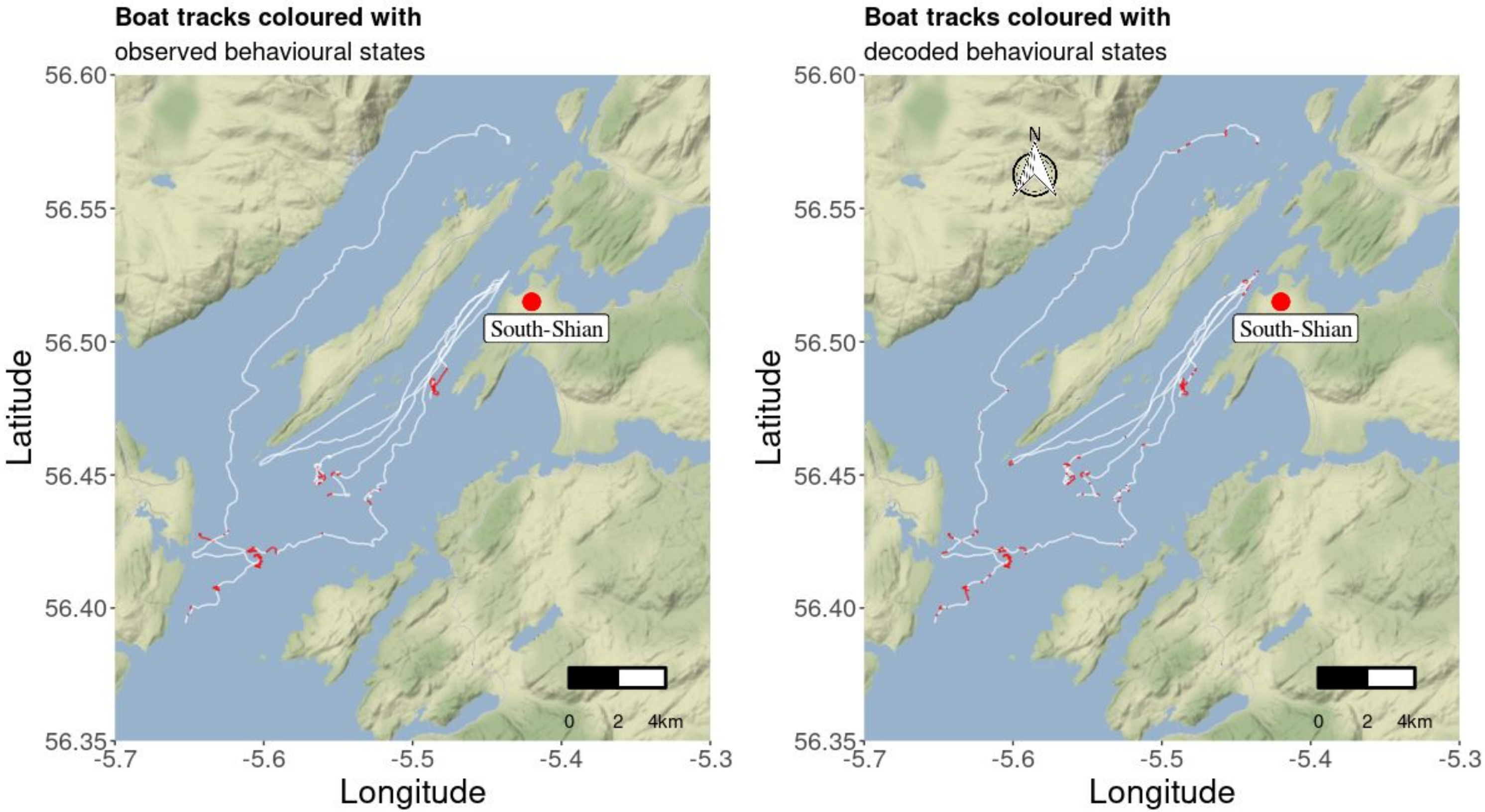
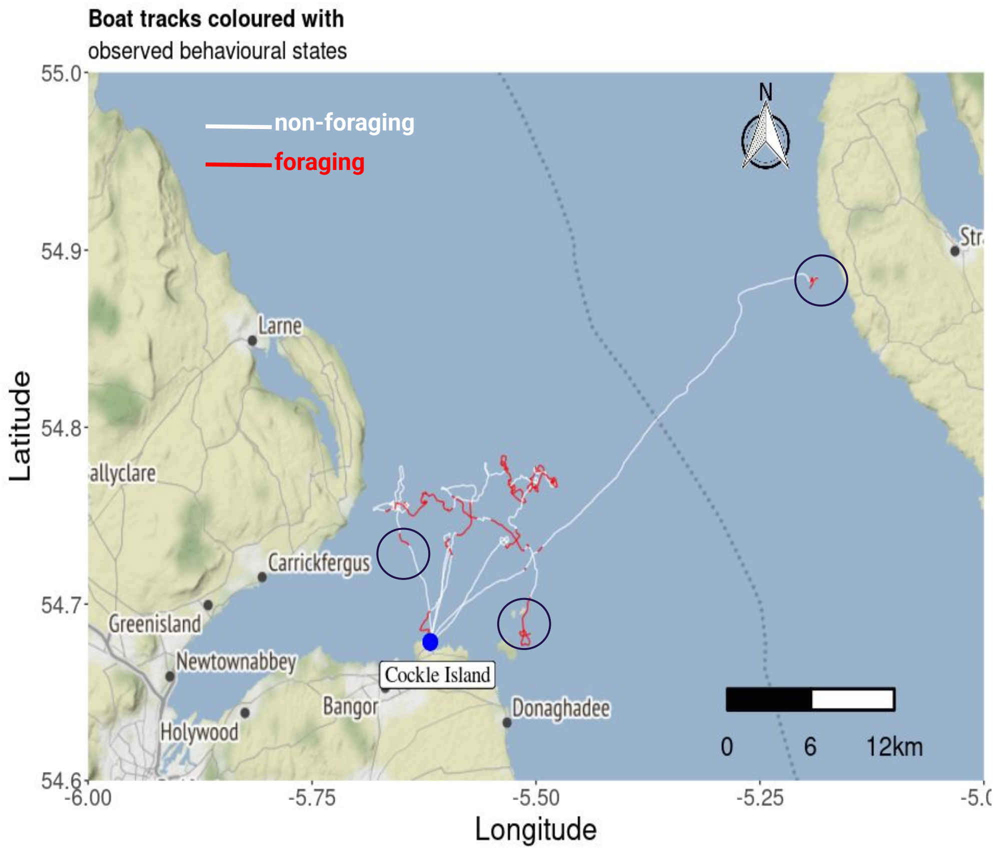


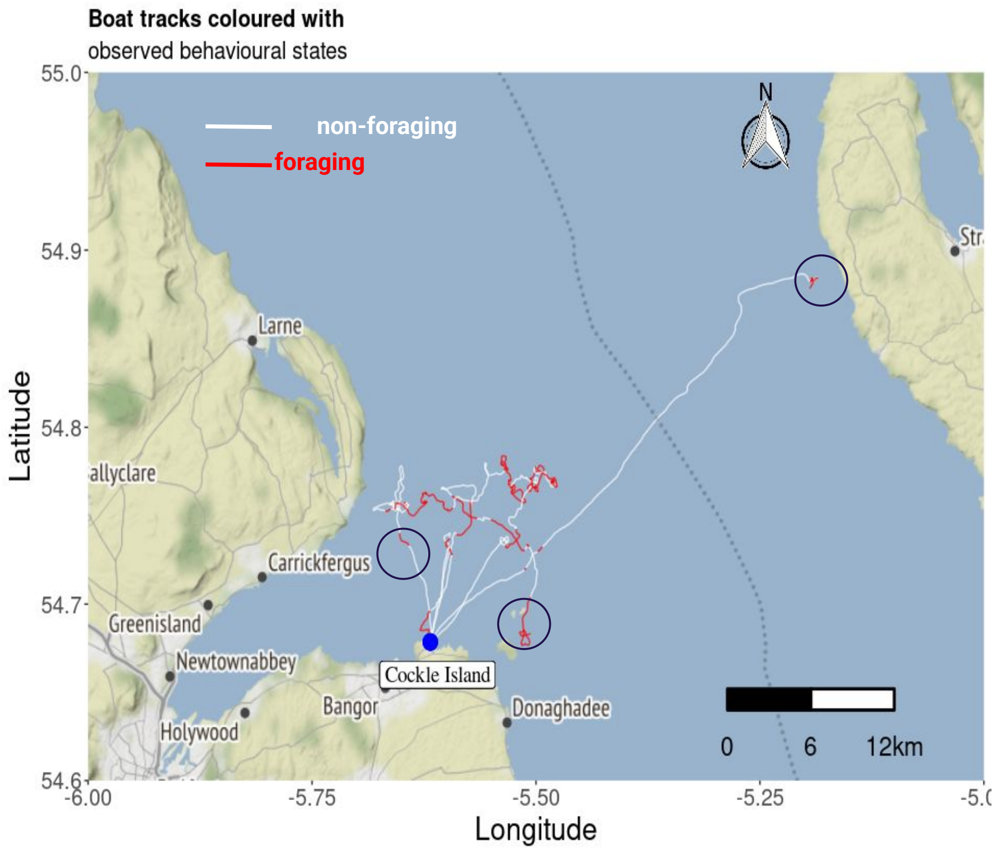
Figure 7: *Visual tracks of common terns coloured with observed (left) and inferred (right) behavioural states: foraging (red) and non-foraging (white), during chick-rearing period, 2011.*

FORAGING EVENTS



- A **foraging event** is a bout within the tracking data where foraging activity occur.

FORAGING EVENTS



- A **foraging event** is a bout within the tracking data where foraging activity occur.

Foraging Events - Chick Rearing

- Inferred foraging states are from HMMs deemed optimal based on validation.

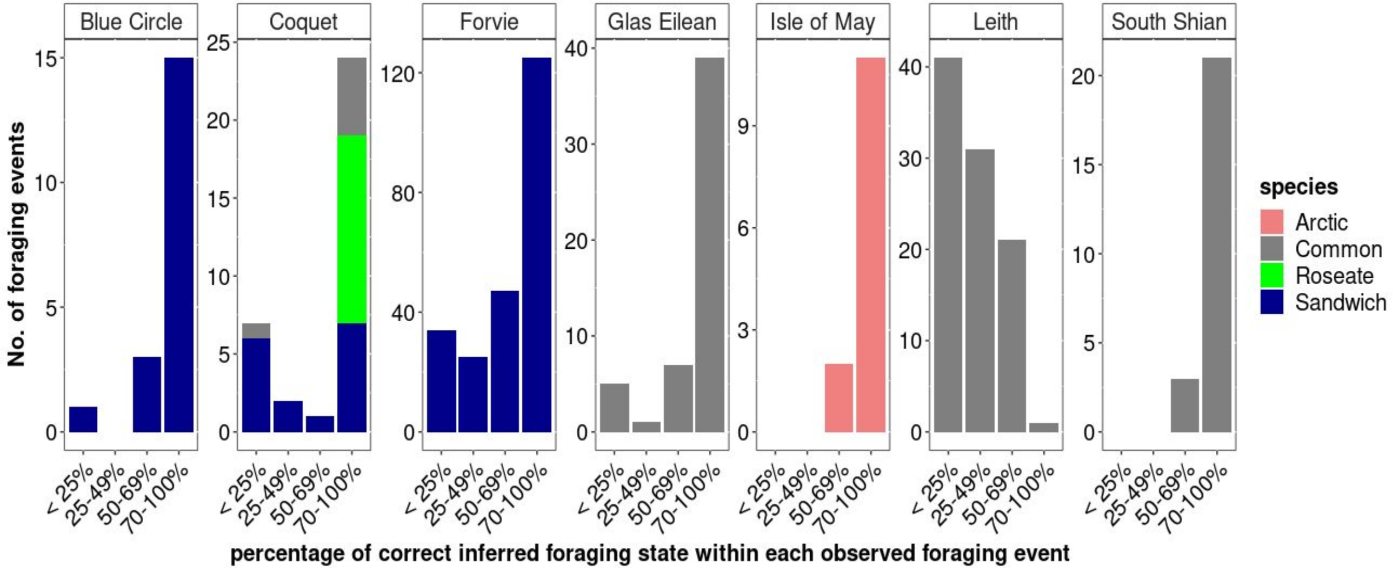


Figure 8: *Proportion of correctly inferred foraging states within each observed foraging event across breeding colonies*

FORAGING EVENTS - INCUBATION

- Inferred foraging states are from HMMs deemed optimal based on validation.

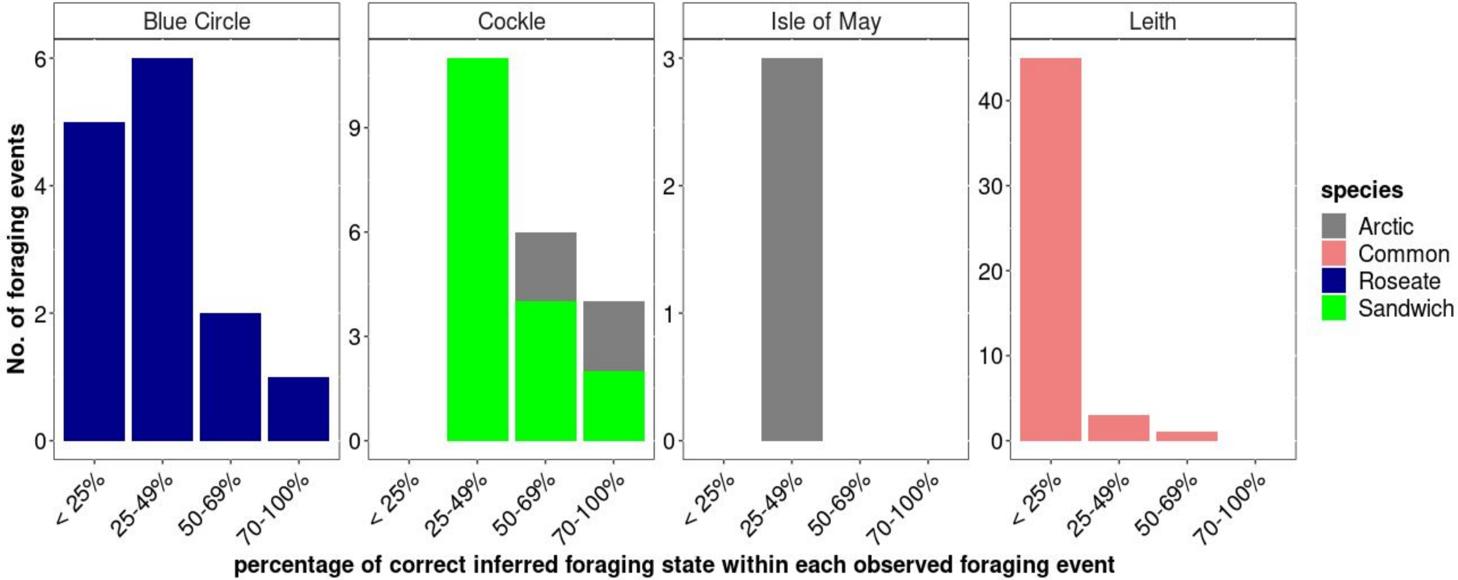
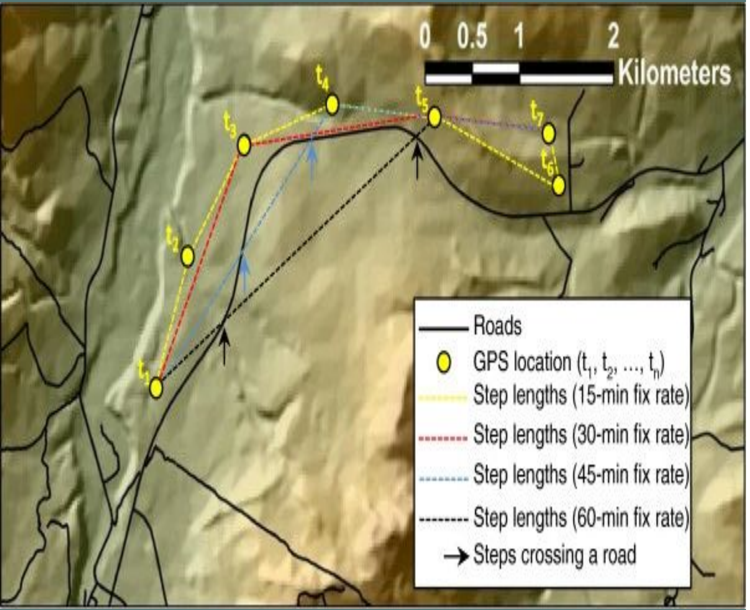


Figure 9: *Proportion of correctly inferred foraging states within each observed foraging event across breeding colonies*

PROBLEM 2



Thurfjell et al. (2014)

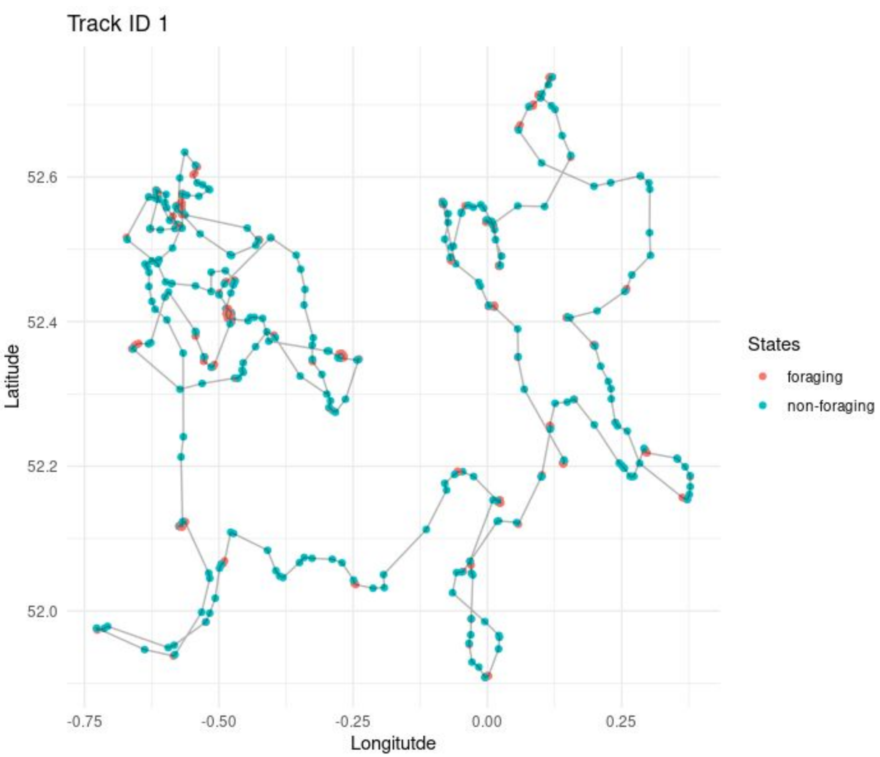
Sampling frequency determines the temporal resolution at which behavioural inferences can be made.

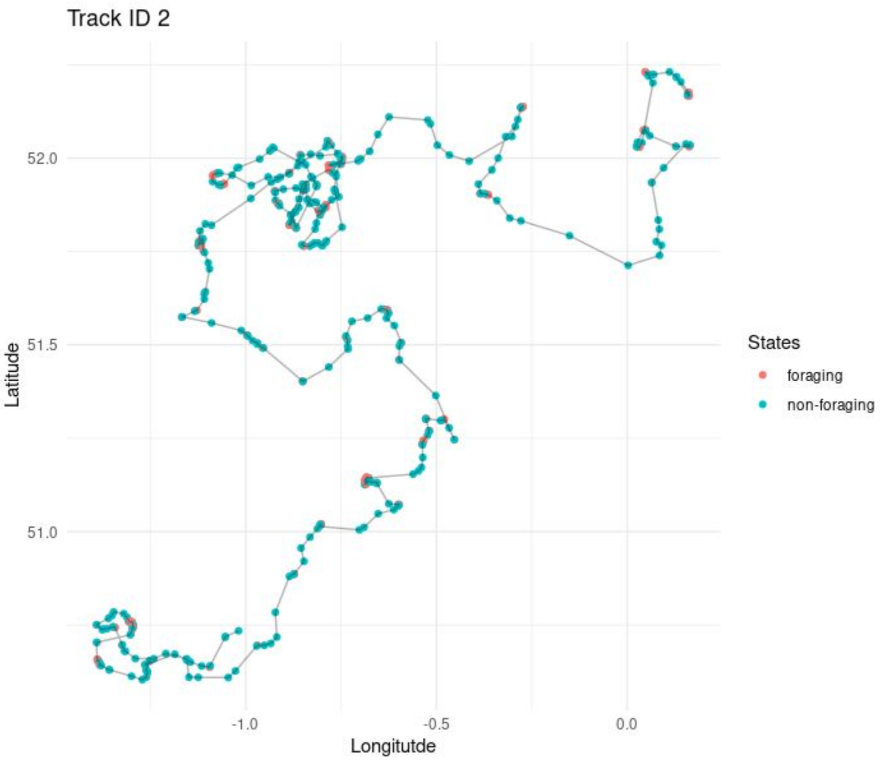
High sampling frequency comes at a cost sampling duration and GPS battery life

Lower sampling frequency may not capture behaviour obtainable at finer temporal scale

Work Flow

Simulate tracking data





Work Flow (Ongoing)

- Resample simulated tracking data
- Investigate the effect of sampling frequency on behavioural state inference
- Application to real life movement data

REFERENCES

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THANKS
FOR
LISTENING

QUESTIONS?